

Forward and inverse processes in model generation

IA pour la Science

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Two aspects of the SCM inference were discussed the last week:

1. The inference avoids the Integer Linear Programming problem by replacing the probabilities of edges

$$\mathbf{\Gamma} \in [0, 1]^{n \times n} \mapsto \mathbf{Z} \in \{0, 1\}^{n \times n}.$$

for the adjacency of edges. *NB: it is solved below.*

2. The list of variables is not fixed. For the set of observable variables $X_1, \dots, X_n$ the model has to forecast the list of variables $X_1, \dots, X_n, \dots, X_{n'}$. By default, the value $n' = 0$. *NB: it is under consideration.*

The foundation model includes the elements of SCM selection:

- 1) maximizing likelihood of the DAG at each step,
- 2) do-intervention actions each step on the sampled node.

The foundation model for the SCM inference

The main type of the foundation model carries the (Riemannian) diffusion process.

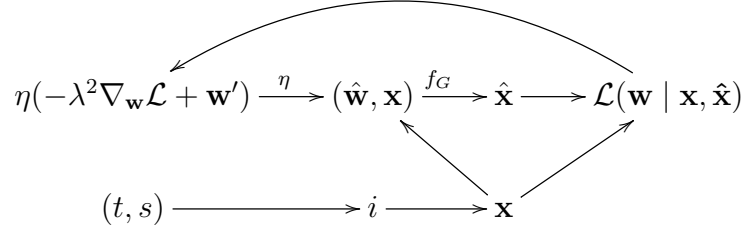
The diffusion process involves the step t and two distributions to approximate the distribution of the adjacency matrix $\mathbf{Z}$. Use the vector form $\mathbf{z} = \text{vec}(\mathbf{Z})$. Denote the training distribution $q(\mathbf{z})$ and the sampling distribution $p(\mathbf{z})$. During the model training one has to minimize the distance between two distributions at each step.

The diffusion process invokes the local structural modeling:

$$\mathbf{Z} \mapsto f_g \mapsto \eta \mapsto \mathbf{w},$$

with η is the optimization algorithm that delivers the optimal parameters to the SCM at each step of the diffusion process.

The *fine-tuning* algorithm η optimizes the parameters according to the diagram



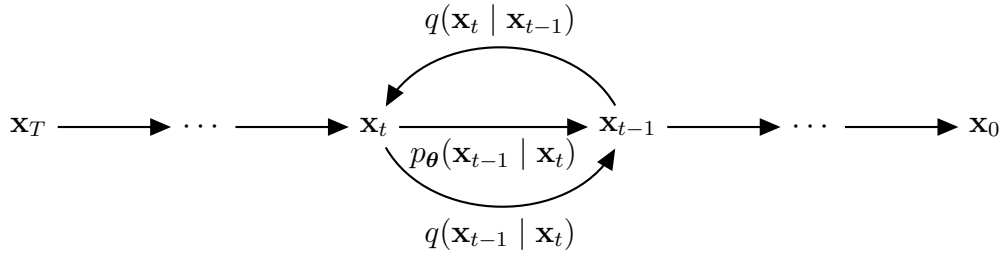
This diagram reads: the optimization algorithm η estimates the SCM parameters $\mathbf{w}$ by the gradient of its loss and the previous parameters to solve the problem

$$\hat{\mathbf{w}} = \arg \min \mathcal{L}(\mathbf{w}) \quad \text{given} \quad f_G(\hat{\mathbf{w}}, \mathbf{x}) \mapsto \hat{\mathbf{x}}.$$

The algorithm η uses some initial value $\mathbf{w}_0$ and a stop-criterion for its iterations.

A basic diffusion process for the data

The data $\mathbf{x}$ only, without the binary adjacent $\mathbf{z}$ for the model f_G are sampled and inferred with the generative process below.



Given the data distribution $\mathbf{x}_0 \sim q(\mathbf{x})$ assert:

- 1) forward diffusion process $\mathbf{x}_t = \sqrt{1 - \beta_t} \mathbf{x}_{t-1} + \sqrt{\beta_t} \mathbf{z}_t$, sampling i.i.d.

$$\mathbf{x}_1, \dots, \mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

- 2) sampling $q(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ and learning the parameters θ of $f_{\text{ai}}: p_{\theta}(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$,
- 3) reverse diffusion process

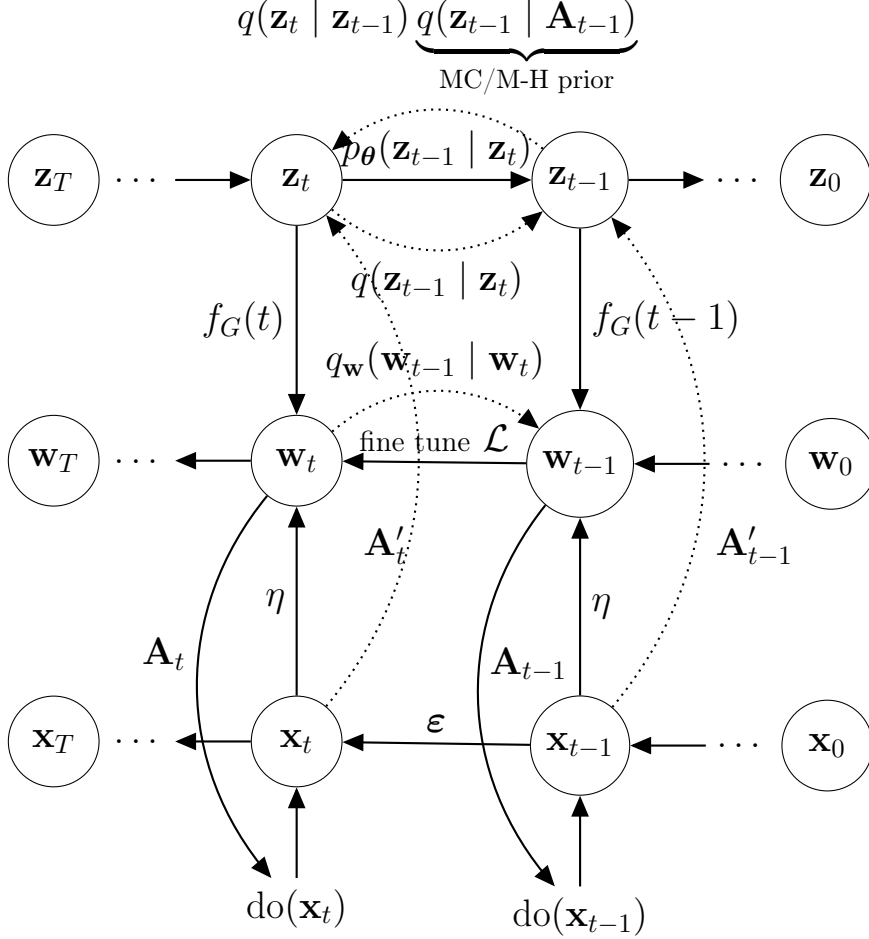
$$p_{\theta}(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^T p_{\theta}(\mathbf{x}_{t-1} \mid \mathbf{x}_t),$$

- 4) slow learning gives

$$p_{\theta}(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_{\theta}(\mathbf{x}_t, t), \boldsymbol{\Sigma}_{\theta}(\mathbf{x}_t, t)).$$

Coherent generation of SCM submodels with do-intervention

Each step of the diffusion delivers a set of submodels to reconstruct. The dotted lines of the diagram are sampling; the rest are inference or optimization.



The algorithm of SCM inference includes the U-Net $f_{\text{ai}}(\boldsymbol{\theta})(\mathbf{x}, \mathbf{z})$ to learn its parameters $\boldsymbol{\theta}$ and make an inference of the SCM f_G along with its parameters $\mathbf{w}$. The estimated parameters $\hat{\mathbf{w}}$ are inherited at each step t . So that only fine-tuning optimization η goes to minimize $\mathcal{L}(\mathbf{w}_{t-1}) \mapsto \mathbf{w}_t$. (The dataset refers to the structure $(\mathbf{X}, \mathbf{z})$ with a transformer.)

1. Sample a new SCM adjacency $\mathbf{z}_t$.
2. Construct the new model f_G and its initial parameters $\mathbf{w}_t$ from the step $t-1$.
3. Fine-tune the SCM parameters $\hat{\mathbf{w}}_t = \arg \min \mathcal{L}(\mathbf{w}_t | \hat{\mathbf{w}}_{t-1})$ with the initial parameters.
4. Estimate the covairiance $\hat{\mathbf{A}}^{-1} = \mathbf{cov}(\mathbf{w})$ using the hessian of $\mathcal{L}$, the Fisher matrix.

5. Apply the CI-criterion $\rho(\mathbf{A}_t)$ to make the intervention $\text{do}(\mathbf{x}_t)$.
6. Make the intervention, sampling $\text{do}(\mathbf{x}_t)$ with the intervention parameters $\mathbf{A}'_t$
7. Compute the partial residuals $\mathbf{x}_{t-1} - \hat{\mathbf{x}}_{t-1} = \boldsymbol{\varepsilon}_t \mapsto \mathbf{x}_t$.
8. Learn the f_{ai} parameters by reconstructing $p_{\boldsymbol{\theta}}(\mathbf{z}_{t-1} \mid \mathbf{z}_t)$.
9. Go to the step 1 and sample the next adjacency $\mathbf{z}_{t-1}$.

List of the modules

The empirical distribution of the observations

$P(X_1, \dots, X_n)$ is the distribution of the observable data. Its approximation is given by the Gaussian Process Regression model

$$p(\mathbf{x} \mid f_{\text{gp}})$$

on the dataset

$$\{\mathbf{x}_i \in \mathbb{R}^n\}.$$

It is indexed by some function of the time and space $(t, s) \mapsto i$.

A model from the family of universal approximators

For a sample $\mathbf{x}$ and its reconstruction $\hat{\mathbf{x}}$ the likelihood function provided a universal model $f_{\text{nn}}(\boldsymbol{\theta}, \mathbf{x})$ is

$$p(\mathbf{x} \mid f_{\text{nn}}, \boldsymbol{\theta}).$$

Without the a priori information about the model parameters $\boldsymbol{\theta}$ the corresponding loss function is

$$\mathcal{L}(\boldsymbol{\theta}) = -\log p(\mathbf{x} \mid f_{\text{nn}}, \boldsymbol{\theta}) = (\hat{\mathbf{x}} - \mathbf{x})^\top \mathbf{B}^{-1} (\hat{\mathbf{x}} - \mathbf{x}).$$

The matrix $\mathbf{B}$ is the covariance of the reconstruction error. This matrix in some statements depends on the index i . This model as a parametric one has a structure to make a distillation to the causal graph

$$f_{\text{nn}}(\boldsymbol{\theta}, \mathbf{x}) \xrightarrow{\text{extract}} \mathbf{Z}_{\text{nn}} \xrightarrow{\text{dist}} \mathbf{Z}.$$

An SCM we have to forecast

A structural causal model

$$\hat{\mathbf{x}} = f_G(\mathbf{w}, \mathbf{x}) \in \mathcal{F}_{\text{scn}}$$

has a structure $\mathbf{Z}$ and its probabilistic representation $\mathbf{\Gamma}$. The elements of the matrix $\mathbf{\Gamma}$ are the probabilities of the causal relationships between the elements of the model structure f_G . The matrix $\mathbf{Z}$ is the adjacency matrix of the causal graph, the binary representation of $\mathbf{\Gamma}$.

Note: *we suppose that SCM does not fit the whole (s, t) -data but a local part of it. So we also call SCM a local model f_G , in contrast to the universal model f_{nn} .*

A foundational model to forecast the SCM

There given pairs a sample $\mathbf{x}_i$ and the corresponding SCM f_G with its structure $\{\mathbf{Z}_i\}$. Call the set of pairs

$$(\mathbf{x}_i, \mathbf{Z}_i), \quad i \in \mathcal{B}$$

a collection of the SCM forecasting. This collection $\mathcal{B}$ shows one case of mathematical modeling that results in an interpretable model f_G . This model fits samples $\mathbf{x}_i$ from some local data $i \in \mathcal{B}$.

Call a model that forecasts an SCM $\mathcal{F}$ a *foundation model*

$$f_{\text{ai}} : \mathbf{x}_{\mathcal{B}} \mapsto f_G \in \mathcal{F}.$$

The SCM is a parametric model

$$f_G = f_G(\mathbf{w}, \mathbf{x}) \quad \text{with the parameters} \quad \hat{\mathbf{w}} = \arg \min_{\mathbf{w}} \mathcal{L}(\mathbf{w}).$$

The foundational model returns not the SCM $f_G(\hat{\mathbf{w}}, \mathbf{x})$ itself, but only a probabilities $\mathbf{\Gamma}$ on the structure $\mathbf{Z}$. So we rewrite the previous definition as

$$f_{\text{ai}} : \mathbf{x}_{\mathcal{B}} \mapsto \mathbf{\Gamma}.$$

Note: *Since a case of mathematical modeling, call it $\mathcal{B}$ has a history of requests to the foundational model, the next information is available for it:*

- 1) *the optimal parameters $\mathbf{w}$ along with $\mathbf{Z}, \mathbf{\Gamma}$ for the SCM f_G ,*
- 2) *the prior probabilities of the model parameters $p(f_G)$ or all models in the family $\mathcal{F}$,*
- 3) *the other information about the case of mathematical modeling $\mathcal{B}$ that is relevant for the forecasting of the SCM.*

The Integer Linear Programming for the causal graph selection

The matrix $\mathbf{\Gamma}$ comes along with the descriptions of its nodes $X_1, \dots, X_n$, and the probabilities. The SCM f_G is defined by the graph $G = (\mathbf{Z}, X)$ by solving

$$\text{ILP}(\mathbf{\Gamma}, P_X) \rightarrow \mathbf{Z} \rightarrow f_G.$$

The evidence of the SCM

A probabilistic description Γ of a model f_G from the class of *admissible models* $\mathcal{F}$ is generated from a parametric distribution

$$\Gamma \sim \text{Dir}(\boldsymbol{\pi}) \quad \text{generated by MCMC} \quad f_{\text{mh}}(\boldsymbol{\pi}),$$

here the lowercase letters mean Metropolis-Hastings Algorithm for MCMC. (A simple genetic algorithm may bring some decent results, too). This distribution provides a probabilistic space for the similar models. See the document on *Multimodeling*.

The model f_G and the corresponding graph has its likelihood

$$p(\mathbf{x}|f_G).$$

See the document on *Causal graph likelihood*.

Note: *Two-level bayesian inference introduces the probability of the SCM parameters on the first level. And the probability of the model on the second level is:*

- 1) *the prior of the SCM is delivered by the generative model f_{mh}*
- 2) *the likelihood of the SCM is delivered by the forecasting model f_{ai} .*

The criterion of causality

The parameters of the model f_G has their covariance matrix. The prior of the model parameters is

$$-\ln p(\mathbf{w}|f_G) = (\mathbf{w} - \boldsymbol{\mu})^T \mathbf{A}^{-1} (\mathbf{w} - \boldsymbol{\mu}).$$

The algorithm of the Bayesian hyperparameters $(\boldsymbol{\mu}, \mathbf{A})$ estimation (and their connection between the causality) is described in the document on *parameter analysis*.

A model for the intervention

See the document on the *intervention strategy*.